

Anshul Pundhir

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OBJECTIVE

I am a Ph.D. candidate at the Machine Intelligence Lab, IIT Roorkee, under the supervision of Prof. Balasubramanian Raman. My research primarily focuses on Medical Image Analysis using Deep Learning techniques, with additional exploration into Federated Learning, Differential Privacy, Remote Sensing, and Spiking Neural Networks. I am committed to learning new skills and collaborating with fellow researchers to tackle complex challenges. I am enthusiastic about making meaningful contributions and seeking postdoctoral opportunities.

EDUCATION

Ph.D. , Computer Science, CGPA: 9.25/10 Indian Institute of Technology, Roorkee, Uttarakhand, India. Thesis Area: <i>Utility Enhancement and Bias Mitigation in Skin Lesion Analysis</i> Supervisor: Prof. Balasubramanian Raman <i>Thesis submitted: 12 July 2024.</i>	Dec 2019 - July 2024
Integrated M.Tech. , Computer Science, CGPA: 8.51/10 Dayalbagh Educational Institute, Agra, Uttar Pradesh, India.	2019
Integrated M.Sc. , Computer Science, CGPA: 8.30/10 Dayalbagh Educational Institute, Agra, Uttar Pradesh, India.	2018
B.Sc. , Computer Science, CGPA: 8.46/10 Dayalbagh Educational Institute, Agra, Uttar Pradesh, India.	2017

ACADEMIC EXPERIENCE

Teaching Assistant <i>Indian Institute of Technology, Roorkee, India.</i> Assisted Prof. Balasubramanian Raman in the tutorial and quiz sessions for Advance Algorithms (CSN-501), Data Structures and Algorithms (AID-525), Fundamentals of Object Oriented Programming (CSN-103), and Discrete Structures (CSN-106) courses, Data Structures (CSC-102), Programming with C and C++ (CSC-101), etc.	Dec 2019 - Present
Teaching Assistant <i>Dayalbagh Educational Institute, Agra, Uttar Pradesh, India.</i> Assisted Dr. R.S. Pavithr in lab and quiz sessions for Soft-Computing and Programming courses.	Jul 2016 - Jul 2018

RESEARCH INTERESTS

- Medical Image Analysis
- Deep Learning
- Machine Learning
- Federated Learning
- Remote Sensing
- Differential Privacy
- Evolutionary Algorithms

PUBLICATIONS

Journals

- [J-1] Kishore Babu Nampalle, **Anshul Pundhir**, Pushpamanjari Ramesh Jupudi and Balasubramanian Raman, Towards Improved U-Net for Efficient Skin Lesion Segmentation, Journal of Multimedia Tools and Applications (MTAP), Springer, 2024 (**Status: Published**).
- [J-2] **Anshul Pundhir**, Shivam Sagar, Pradeep Singh, and Balasubramanian Raman, “Echoes of Images: Multi-loss Network for Image Retrieval in Vision Transformers”, published in the Journal of Medical & Biological Engineering & Computing (MBEC), 2024 (**Status: Published**).
- [J-3] **Anshul Pundhir**, Balasubramanian Raman, Pravendra Singh “Biasing & Debiasing based Approach Towards Fair Knowledge Transfer for Equitable Skin Analysis”, Journal of Biomedical Signal Processing and Control (BSPC), 2024 (**Status: Under Review**).
- [J-4] **Anshul Pundhir**, Balasubramanian Raman, Pravendra Singh “Intra-Class Interaction based Co-Correcting Dual Branch Framework for Fair Skin Lesion Analysis”, Journal of Neurocomputing, 2024 (**Status: Under Review**).

Conferences

- [C-1] **Anshul Pundhir**, Srishti Yadav, Balasubramanian Raman, Sanjeev Kumar, “Unveiling Robustness of Spiking Neural Networks Against Data Poisoning Attacks”, International Joint Conference on Neural Networks (IJCNN), 30 June - 5 July 2024, Yokohama, Japan (**accepted for publication**).
- [C-2] **Anshul Pundhir**, Sanchit Verma, Balasubramanian Raman, “Towards Ethical Dermatology: Mitigating Bias in Skin Condition Classification”, International Joint Conference on Neural Networks (IJCNN), 30 June - 5 July 2024, Yokohama, Japan (**accepted for publication**).
- [C-3] Srishti Yadav, **Anshul Pundhir**, Tanish Goyal, Balasubramanian Raman, Sanjeev Kumar “Differentially Private Spiking Variational Autoencoder”, International Conference on Pattern Recognition (ICPR), Dec 01-05, 2024, Kolkata, India (**accepted for publication**).
- [C-4] **Anshul Pundhir**, Saurabh Dadhich, Ananya Agarwal, and Balasubramanian Raman, “Metadata guided Visual Attention Fusion for Improved Skin Lesion Classification using Deep Learning”, 12th workshop on Multimodal Learning and Fusion Across Scales for Clinical Decision Support (ML-CDS) in conjunction with MICCAI, September 18, 2022, Singapore (**published**).
- [C-5] **Anshul Pundhir**, Saurabh Dadhich, Ananya Agarwal, and Balasubramanian Raman, “Towards Improved Skin Lesion Classification using Metadata Supervision”, 26th International

Conference on Pattern Recognition (ICPR 2022), August 21-25, 2022, Montréal Québec, Canada (**published**).

- [C-6] **Anshul Pundhir**, Deepak Verma, Puneet Kumar, and Balasubramanian Raman. “Region extraction based approach for cigarette usage classification using deep learning”, 6th IAPR International Conference on Computer Vision and Image Processing (CVIP), December 3-5, 2021 IIT Ropar, Punjab-140001, India (**published**).

PATENT FILED

1. Pravendra Singh, **Anshul Pundhir**, and Balasubramanian Raman, “System and Method for Enhancing Fairness in Imaging Applications through Mutual Correction and Groupwise Intra-Class Sample Interaction”, Status: **Filed**, Filing No. 202411037096, Filed on 10/05/2024 (Funded by IIT Roorkee).

AWARDS & SCHOLARSHIPS

1. **Registration support** from the **Dean of Resources and Alumni Affairs (DORA)**, IIT Roorkee to present a research paper in 26th International Conference on Pattern Recognition (ICPR 2022), August 21-25, 2022, Montreal Quebec, Canada
2. **Registration support** from the **Microsoft Research Grant** from the Department of Computer Science and Engineering IIT Roorkee to present a research paper in 12th workshop on Multimodal Learning and Fusion Across Scales for Clinical Decision Support (ML-CDS) in conjunction with MICCAI, September 18, 2022, Singapore
3. **Registration support** from the **UGC-JRF Contingency fund** to present a research paper in the 6th IAPR International Conference on Computer Vision and Image Processing (CVIP 2021), **IIT Ropar, India**
4. Received **UGC-JRF Scholarship** during Ph.D. (Dec, 2019 onwards)
5. Gave a topper talk on chamomile tea with toppers for JRF (available on YouTube) [Link]
6. Qualified **NTA NET-JRF June with 99.88%ile** Computer Science in 2019
7. Received **MHRD Scholarships** during M.Tech in 2019
8. Qualified **NTA NET June, Dec** Computer Science in 2018
9. Qualified **GATE** Computer Science in 2018
10. Awarded third in National Science Exhibition
11. Third rank in city, National Science Olympiad (NSO)

TECHNICAL SKILLS

Programming Languages	: C/C++ (proficient), Python (proficient), Core Java (acquainted).
Development Tools	: MATLAB, PyTorch, Keras, Tensorflow, Anaconda, PyCharm, Spyder, Eclipse, Visual Studio Code, Git and GitHub, LaTeX, Docker.
Operating System	: Ubuntu (proficient), Windows (proficient).

OTHER PROJECTS

- *Video Analytics*
Application of computer vision algorithms for academic purpose systems.

- *Intrusion Detection System*
Use feature elimination technique to get relevant features for system.
- *Credit Card Fraud Detection System*
Detect fraudulent transactions using ML techniques.
- *Twitter Sentiment Analysis*
Analyze tweets to perform sentiment analysis.
- *Automated Essay Grader*
Use NLP to evaluate the effectiveness of the given essay.
- *Time-Table Utility Software*
Extract information from given time table schedule.

WORKSHOP/TRAINING/CONFERENCES ATTENDED

1. Six-months Deloitte work-studentship program from IIT Roorkee (November 2023 - April 2024), 2023.
2. Research Internship at Samsung R&D Institute Bangalore (SRI-B) India (16 August 2022 - 17 February 2023), 2022.
3. Attended 12th workshop on Multimodal Learning and Fusion Across Scales for Clinical Decision Support (ML-CDS) in conjunction with MICCAI, September 18, 2022, Singapore, 2022.
4. Attended 26th International Conference on Pattern Recognition (ICPR 2022), August 21-25, 2022, Montreal Quebec, Canada, (Virtual) 2022.
5. Attended 6th International Conference on Computer Vision and Image Processing (CVIP), IIT Ropar, India (Virtual) 2021
6. Summer internship under the supervision of Prof. Venkatesh Babu from IISc Bangalore (April-July, Virtual) 2021.
7. Attended 6th International Conference on Computer Vision and Image Processing (CVIP), IIT Ropar, India (Virtual) 2021.
8. Attended 28th International Conference on Image Processing (ICIP), Anchorage, Abu Dhabi, UAE (Virtual) 2021.
9. Attended The Workshop on Computer Vision and Image Processing (WCVIP), IIT Roorkee, India (Virtual) 2020.

TALKS/WORKSHOPS DELIVERED

1. Conducted a training session on ‘AI Applications in Healthcare’, in Executive Certificate Program in AI Strategy for Leaders, iHUB DivyaSampark, TIH, IIT Roorkee with The Economic Times, India (Virtual), July 2022.
2. Conducted a training session on ‘Supervised and Unsupervised Machine Learning Algorithms’, in Executive Certificate Program in AI Strategy for Leaders, iHUB DivyaSampark, TIH, IIT Roorkee with The Economic Times, India (Virtual), June 2022.
3. Assisted a Faculty Development Program (FDP) on ‘Machine Learning for Computer Vision (ML4CV)’ at E&ICT Academy, IIT Roorkee, March 2022.

4. Conducted a ‘Practical Session on Clustering Algorithms’ in an online Certificate Course on Applied Data Science and Machine Learning with Python, E&ICT Academy course, IIT Roorkee, India (Virtual), Nov 2021.
5. Conducted a training session on ‘Clustering Algorithms and their use cases’ in an online Certificate Course on Applied Data Science and Machine Learning with Python, E&ICT Academy course, IIT Roorkee (Virtual), Nov 2021.
6. Assisted in a training session on ‘Human Action Recognition and Generation’ during ‘The Workshop On Computer Vision And Image Processing (WCVIP)’ at IIT Roorkee, Dec 2020.
7. Organized a training session on ‘Dimensionality Reduction Algorithms’ during the workshop on Digital Image Processing & Applications (DIPA) at NIT Arunachal Pradesh, Aug 2020.

ACTIVITIES

Research Paper Reviewer: Reviewed research papers in various reputed conferences, including Medical Image Computing and Computer Assisted Intervention Society (**MICCAI**), International Conference on Pattern Recognition (**ICPR**), International Conference on Computer Vision & Image Processing (**CVIP**), International Conference on Neural Information Processing (**ICONIP**), and International Conference On Electrical Electronics Communication and Computers (**ELEX-COM**).

Co-curricular Activities: Volunteer for data collection in a collaborative project between **Michigan State University (MSU), USA** and Dayalbagh Educational Institute, Agra, India, Volunteered in National Service Scheme (NSS), Regular participant in dramatic events in college like Street-plays, skits, Mime, Nukkad Natak, etc.

Hobbies: Watching movies, Hypothetical storytelling, Music, Meditation, and Volunteering.

PERSONAL DETAILS

Father’s Name	: Pushpendra Veer Singh Pundhir	Mother’s Name	: Sadhana Pundhir
Date of Birth	: Aug 13, 1996	Martial Status	: Unmarried
Nationality	: Indian		
Permanent Address: 605, Verma Nagar Agra Road Etah, Uttar Pradesh, India - 207001.			

DECLARATION

I hereby state that all the information mentioned above is true to the best of my knowledge and I shall be held responsible for any discrepancy found later.

Date: August 9, 2024

Place: Roorkee, Uttrakhand.

Anshul Pundhir

REFERENCES

Prof. R. Balasubramanian

Professor, CSE Department, Indian Institute of Technology, Roorkee, India.

E-mail: bala@cs.iitr.ac.in

Prof. Pravendra Singh

Assistant Professor, CSE Department, Indian Institute of Technology, Roorkee, India.

E-mail: pravendra.singh@cs.iitr.ac.in

Prof. Gur Saran Adhar

Professor, Computer Science Department, University of North Carolina (UNCW),
Wilmington.

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Prof. Sandeep Paul

Professor, Physics and Computer Science Department,
Dayalbagh Educational Institute, Agra, Uttar Pradesh, India.

E-mail: spaul@dei.ac.in